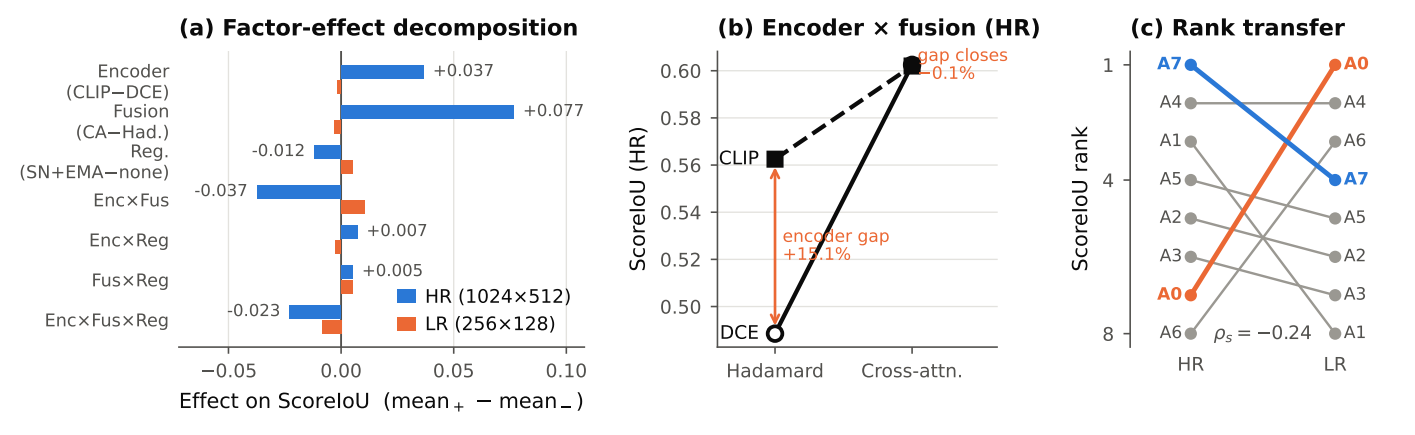
(a) Factor-effect decomposition
Encoder (CLIP−DCE)
Fusion (CA−Had.)
Reg. (SN+EMA−none)
Enc×Fus
Enc×Reg
Fus×Reg
Enc×Fus×Reg
+0.037
+0.077
-0.012
-0.037
+0.007
+0.005
-0.023
HR (1024×512)
LR (256×128)
−0.05
0.00
0.05
0.10
Effect on ScoreIoU (mean+ − mean−)
(b) Encoder × fusion (HR)
ScoreIoU (HR)
0.60
0.58
0.56
0.54
0.52
0.50
CLIP
DCE
gap closes −0.1%
encoder gap +15.1%
Hadamard
Cross-attn.
(c) Rank transfer
ScoreIoU rank
1
4
8
A7
A4
A1
A5
A2
A3
A0
A6
A0
A4
A6
A5
A7
A2
A3
A1
$\rho_s = -0.24$
HR
LR